A DEEP LEARNING-BASED APPROACH FOR EXTRACTION OF INFORMATION FROM ELECTRONIC HEALTH RECORDS

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DECLARATION

I, **Fatima Masood,** D/o **Muhammad Masood**, ID: **MSSE71F23S006,** Session **2023-2025** student of **MS** in **Software Engineering**, hereby declare that the research work entitled **"A Deep Learning-based Approach for Extraction of Information from Electronic Health Records"** presented in this thesis is my original work. It has not been copied, stolen, or plagiarized from any source. Furthermore, this thesis has been prepared in accordance with the prescribed format and guidelines of the university.

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It is certified that the research topic and synopsis have been prepared in accordance with the University's approved format. Furthermore, no prior research has been conducted on a similar topic.

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DEDICATION

I dedicate this thesis to my father, mother, husband, and family, who supported me and gave me the strength to work hard and more efficiently. They are a guiding light for me for this thesis. Moreover, I dedicate this thesis to my supervisor, co-supervisor, and teachers, whose knowledge, encouragement, and support have made this achievement possible.

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LIST OF ABBREVIATIONS

Abbreviations	Stands for
AI	Artificial Intelligence
BiLSTM	Bi-directional Long Short Term Memory
DL	Deep Learning
EE	Event Extraction
EHRs	Electronic Health Records
FN	False Negatives
FP	False positives
GRU	Gated Recurrent Unit
GUI	Graphical User Interface
ICU	Intensive Care Unit
IR	Information Retrieval
IS	Information Structuring
IE	Information Extraction
KBS	Knowledge-Based System
LLMs	Large Language Models
LSTM	Long Short-Term Memory
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
QA	Question Answering
RE	Relation Extraction

RNNs	Recurrent Neural Networks
TN	True Negatives
TP	True positives

LIST OF PUBLICAITONS

ABSTRACT

Information extraction is the automatic extraction of structured data from its unstructured or semi-structured form. Electronic Health Records are important as they contain useful information in unstructured form, so we can apply an information extraction strategy to extract useful structured information from them. There are many traditional ways of information extraction, including Manual and rule-based information extraction. However, rule-based extraction is insufficient and does not cover all the aspects, and manual extraction is effective but slow, expensive, and not scalable. Deep learning is a subset of machine learning to solve complex problems using multilayered neural networks. In recent years, deep learning models have performed very well in different Natural Language Processing tasks. In this study, we intended to use various deep-learning techniques to extract useful information from unstructured Electronic Health Records. It enabled a more precise and efficient information extraction, which helped to make better clinical decisions and enhanced overall healthcare delivery. The key feature of deep learning is to combine the state-of-the-art of Natural Language Processing with neural network architecture for unstructured medical data. Firstly, we preprocessed the EHR data, including tokenization, stop word removal, text normalization, stemming, and lemmatization. Then, we applied deep learning models to extract information. Finally, we validated the extracted information using different post-processing techniques to ensure accuracy and reliability. We measured the results by calculating the Accuracy, Precision, and Recall.

Chapter 1

Introduction

1. Introduction

When digital technologies are evolving, an increasing volume of new information continues to be produced daily. The arrival of social web applications has played a significant role in the rapid growth of unstructured data created spontaneously by users. This phenomenon leads to an enormous amount of unstructured text-based information that is available.

Development in technology is going at an ever-increasing rate, which is causing the massive production of new information every day. The increase in data resulting from the expansions of the social web has led to a large amount of text-based unstructured information. The inability to develop the key insights due to the absence of structure in the data makes it difficult, which requires the creation of efficient Information Extraction systems. Information Extraction (IE) is at the core of Natural Language Processing (NLP) where the focus is to extract and formalize structured information in non-structured document formats. Its main functions include Named Entity Recognition (NER), Event Extraction (EE), and Relation Extraction (RE), which are the components of the architecture of transforming unstructured textual information into structured, practical, and usable, as well as machine-readable information. Many applications, such as Information Retrieval (IR), question answering, knowledge bases, reasoning and decision-making support require such structured formats. Among the various methods of identifying particular clinical entities, one can mention NER (N. Abdelmageed et al., 2022), (K. Jeon et al., 2022), (N. Nath et al., 2022), sentiment analysis (G. D'Aniello et al., 2022), (M. Wankhade et al., 2022), and text classification (A. Mohammed et al., 2022), (Q. Li et al., 2022), (A. Gasparetto et al., 2022) as the different techniques used to identify specific clinical entities (Y. Okamura et al., 2022), assess patients' sentiments and classify clinical data.

The transition from physical health records to automated Electronic Health Records (EHRs) has brought us into a new age of handling patient data, greatly improving storage, accessibility, and the ability to analyze. The EHR contains a wealth of medical information (Chen et al., 2021). EHRs are a full digital repository of clinical data, including demographic info, medical history, prescriptions, test results, remediation suggestions, as well as the notes by the healthcare providers. The continuous generation and management of these documents in healthcare settings generates a massive amount

of data, which is very useful, though not structured. After analyzing this unstructured clinical text effectively, many medical usages are possible, such as diagnosing conditions, clinical decision support, forecasting of patient outcomes, and risk assessment. However, the complexity and variability of medical language, along with the presence of domain-specific terminologies, abbreviations, and contextual dependencies, make the task of information extraction from EHRs particularly challenging. The use of DL for general EHR data analysis is summarized by some recent reviews (C. Xiao et al., 2018), (B. Shickel et al., 2018), (J.R. Ayala Solares et al., 2020), (Y. Si et al., 2021).

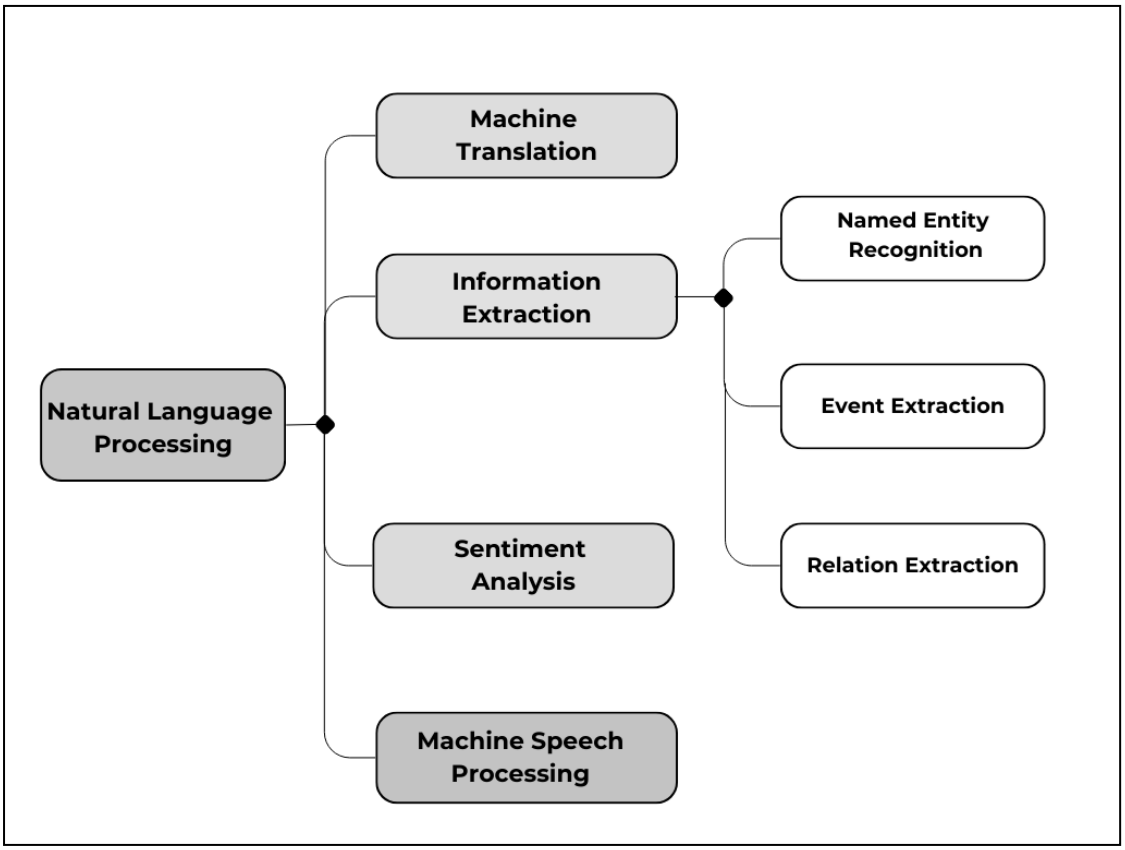

Figure 1. NLP Hierarchy

The first step in the formation of an organized and lexically meaningful medical knowledge graph is the recovery of relevant information from unstructured clinical notes in EHRs. Knowledge graphs give a structured approach to representing medical information that is reusable by healthcare professionals and AI systems to investigate relations between diseases, symptoms, drugs, and the outcomes of patients. By shaping data, hospitals may be in a better position to implement clinical systems for decision support, tailored care plans, predictive models, and automated report creation.

Moreover, the retrieval of structured information from clinical notes facilitates better interoperability of healthcare systems, facilitating easy data exchange and integration amongst medical institutions and research labs.

1.1.1 Challenges in Existing Information Extraction Methods

Different forms of healthcare information are available, such as text, tables, and organized databases. Most of the clinical data in EHRs are written as unstructured free-text narratives and often in shorthand by physicians, thus difficult to automate for extraction. The clinical narrative to structured information process is difficult and has historically relied on manual labor or automated rule-based approaches.

One challenging issue for current information extraction methods is that they rely heavily on manual extraction of medical data. Manual abstraction, despite being called the gold standard, has limitations because human reviewers are not perfect and may be less accurate in some instances (M.N. Zozus et al., 2015), (J.E. Feng et al., 2020), (A.H. Nordo et al., 2017), (T.R. Campion et al., 2010), (R. Brundin-Mather et al., 2018), (D.K. Vawdrey et al., 2007). Healthcare staff or, on a more technical level, the medical coders are generally the ones who examine the records and label them. Although it is highly accurate, it will require a lot of time and human resources, be costly, and present the threat of errors that result from manual processes. The data explosion in healthcare means that manual extraction is unscalable and fails to meet the increasing needs for immediate medical intelligence.

Another shortcoming of current methods for information extraction is the fact that rule-based systems prevail, and data irregularity. Inconsistent data is ubiquitous in temporal EHR (T. Pham et al., 2017), (N.Y. Jane et al., 2016), as the length of a time between different encounters can be inconsistent, leading to difficulties of modeling the entire time series (Xie et al., 2022). According to this approach, clinical text is examined through traditional linguistic rules and pattern-matching algorithms that detect entities and relationships. Although these approaches are useful for accurate assessment on certain occasions, they fail to provide the flexibility that is required to deal with various writing styles, a changing medical vocabulary, and confusing sentence construction. In addition, managing uncertainty as well as the dynamic language is not easy for rule-based systems, which leads to poor recall and performance.

1.1.2. Deep Learning for Information Extraction from EHRs

To overcome the limitations of people-based and rule-based techniques, researchers have pivoted from these techniques and are concentrating on Deep Learning (DL)-based methodologies for information extraction in electronic health records. The state-of-the-art deep learning-based approaches, that is, recurrent neural networks (RNNs) (Z. Che et al., 2018), (H. Hewamalage et al., 2021), long short-term memory (LSTM) (Baytas IM et al., 2017), (G. Maragatham et al., 2019), (W. Lu et al., 2020), and gated recurrent unit (GRU) (Khoshnevisan F et al., 2018), represent state-of-the-art because of the drawbacks of the standard learning algorithms in addressing challenges. This pair of sequential deep learning structures is possibly applicable to address this temporal characteristic of the EHR. The deep learning algorithms display their superiority in temporal EHR data modeling abilities through their learning ability, flexibility, and the generalizability due to non-linearity by using complex non-linearity (H.J. Park et al., 2020), (B.K. Reddy et al., 2018), (Yang Y et al., 2017), (Hung C et al., 2017). Systems that belong to such a broader field of Machine Learning, as deep learning models, use layered artificial neural networks to identify the signs and semantic structures in the big data sets. Transformers, instead of models like BERT and its specialization, BioBERT, have been doing remarkably well in NLP fields such as NER, RE, and text classification.

By introducing deep learning, the automatic information extraction process could avoid the formulation of explicit rules, instead allowing models to obtain insights from annotated datasets straightforwardly. These methods rely on pre-training language models, sequence tagging for NER, and GNNs for RE. Translating big transformer networks to medical language sources increases their efficiency in being able to understand terminology and split complex sentences in medical terms. Modern models such as Bi-directional Long Short Term Memory BiLSTM-CRF and transformer-based classifiers are used, regularly, to recognize medical entities such as diseases, medications, or symptoms. Through the examination of relationships between entities, GNN models can produce clinically appropriate, robust, and pertinent information.

The use of Deep learning to extract information from EHRs enables us to reduce the manual workload, increase scalability, and additionally increase the accuracy of structured data extraction. This makes knowledge discovery more rapid and ensures that healthcare is evidence-based.

1.1.3. Significance of This Study

The growing adoption of Artificial Intelligence (AI) in healthcare has increased the need for accurate, automated extraction of structured information from medical texts. The potential benefits of an efficient IE system for EHRs include:

- **Improved Clinical Decision Support**: Necessary patient data are able to be retrieved faster, and, consequently, physicians are able to make more suitable treatment choices.
- **Enhanced Predictive Analytics:** When properly implemented, the techniques of the Machine Learning may estimate the development trajectory of the disease, identify risks that the patients could have and manage resources more efficiently.
- **Facilitated Medical Research:** This model can be applied to examine vast clinical datasets in order to elicit information on trends in diseases and the general efficacy of treatment procedures.
- **Interoperability in Healthcare Systems:** Due to the standardized data extraction, hospitals and medical databases can easily communicate and cooperate with one another.

1.1.4. Proposed Approach

In this study, we intend to design a deep learning-based system of information extraction from unstructured EHRs. The proposed framework successfully identifies clinical entities and relationship among medical documents with the assistance of the NLP technologies. Our approach is more precise and has better generalization ability with different datasets using knowledge-based model combined with deep learning techniques. Design and execution of a scalable solution with real-time automated information extraction in real-life clinical practice.

IE is important in the process of transforming unstructured text data into structured data to facilitate numerous applications across multiple disciplines. In healthcare, the extraction of structured information from EHRs is essential for helping make clinical decisions, predict diseases, and enhance medical research. However, manual annotation and rule-based extraction are limited in scalability, efficiency, and accuracy. Deep learning-based solutions promise a lot, thanks to their capability to automate the

extraction process, train the systems on large sets of data, and produce enhanced accuracy in the retrieval of structured information. Data were extracted using automated processing and included the data of all patients tested on SARS-CoV-2 or diagnosed with COVID-19 within EHRs during the study period (Yin et al. 2022).

The following study aims to formulate an automated deep learning solution for EHR data extraction, aiming to eliminate current limitations and move forward with data-driven healthcare analytics and decision support. Making use of advanced NLP technologies, this research seeks to advance AI healthcare applications, enabling the use of patient data and helping to achieve more effective health outcomes.

1.2. Background

Progress in digital healthcare technology has led to an explosive growth in EHRs, and most of the medical data is now stored in an unstructured manner. Such basic information about patients includes clinical notes, diagnoses, prescription records, plan of treatments, and even demographics are documented within these papers. Picking out important information from EHRs is essential to making clinical decisions, predicting data analytics, and empowering medical research. Even though EHRs are complex, there is also a need for elaborate strategies to ensure that unformatted text can be formatted into a usable set of structured data.

The existing basic methods of extracting valuable information from EHRs are more likely to be based on rule-based process, machine-learning models, and the use of NLP. These systems are based on the pre-existing conventions and a hand-made template that is used by medical practitioners to identify the medical entities. These are easy to work under controlled environments, but normally do not scale and need edits to reflect new medical terms and language as new languages emerge. The rule-based systems as well are not robust considering that they rely on variations in sentences and application of clinical terminologies on different health settings.

A significant amount of research has been conducted that has centered on supervised machine learning as an aid to extract and detect medical entities in clinical documents. Using the example data in capitalizing on entity extraction, the models maximize pattern recognition, and enable identification of key elements in medicine. Nevertheless, it is slow to acquire the data that is to be labeled. Conventional ways of

machine learning are also not able to interpret new data and this implies their inability to be applied equally to different institutions and patients in the medical field.

Medical information extraction has altered with the new technological advancements in NLP and deep learning. Other deep learning models that use transformers such as BERT, BioBERT, and Med-BERT have shown immense advances in organizing data within the unstructured environment of the EHR. Using the complex patterns of language and context, these models can facilitate the identification of named entities and the identification of relations, as well as multi-label classification, which is complete. In contrast to the rule-based and traditional machine learning models, DL models may automatically learn hierarchical representations of raw text, thereby eliminating much feature engineering and manual rule creation requirements.

Although huge improvements have been achieved, clinically accessed EHRs still create difficulties in retrieving the clinical information because of vagueness, poly semantic forms, and contextual peculiarities in the medical records. The presence of professional medical language, short-hand idiom and ungrammatical constructions in the clinical texts makes it very frustrating to make an effective extraction attempt. Specifically, the fact that various interpretations will be allowed regarding such terms as “stress” (one of them being mental and the other one being physical) underlines the necessity to have the methods of effective disambiguation.

Unique perspectives, including combining deep learning strategies with knowledge bases, promise a lot in overcoming these challenges. The integration of deep learning algorithms in conjunction with ontologies about specific areas (e.g., UMLS and SNOMED CT) brings information extraction better context and improved discrimination of similar terms. In addition, the incorporation of multi-task learning strategies facilitates simultaneous entity and relationship and event detection, hence increasing both processing efficiency and precision.

A major aspect of medical information extraction is RE, to identify connections between medical entities (drug-disease connections, symptom-diagnosis relations). Traditionally, RE is handled with the help of dependency parsing and co-occurrence detection; however, the development of new methods of deep learning, such as GNNs and attention mechanisms, has considerably improved the accuracy of relation extraction. It will be easier to provide better clinical decision aid and benefit medical

research through the improvement of data utilization in the automation of knowledge graph generation for medical applications using these approaches.

The emerging body of work has investigated various ways of obtaining valuable information from EHRs. Experimental results show that the models based on deep learning highly enhance precision and recall results, especially when dealing with large-scale datasets from clinical records. However, important challenges such as protecting the privacy of patients, improving model transparency, and dealing with computational requirements remain major concerns for researchers. Crossing the line on the ethical and secure use of patient data is key, and this calls for the intent of deploying federated learning and privacy-preserving mechanisms.

Generally, traditional methods underpin the medical information extraction, yet deep learning-based solutions offer enhanced reliability and scalability in this field. To cater to the need for automated mining of structured data in EHRs, the present study aims to develop an advanced IE system with the use of advanced NLP tools, domain-specific resources, and multi-task learning techniques. The approach under development is set to strengthen clinical decision support, improve patient care, and add value to research in medical informatics.

1.3. Research Problem

EHRs are filled with many of important clinical data in the form of unstructured text of existing physicians' notes, histories of the patients, prescriptions, and diagnostic findings. These records could be invaluable to clinical decision making, medical research as well as healthcare analytics, but due to the unstructured format of these records, effective automated processing and information retrieval is challenging. Extraction of the important clinical data from EHRs at the normal practice is hectic and prone to errors and it is increasingly becoming difficult as additional patient data is produced. This inefficient efficiency fails to enable clinicians, researchers, and data scientists to exhaustively exploit such insights to drive the improvement of patient welfare, medical decision support, and medical knowledge development.

The current information extraction processes in healthcare mainly agree with the rule-based process, word search and statistical model. These systems can be understood, and they are costly to build and maintain and can only be utilized in a limited number of healthcare facilities. Analogous endeavors using conventional NLP

approaches, such as NER and parsing of dependencies, present complexities in accurately mapping complicated medical jargon, abbreviations, and relationships that require contextualization. Though machine learning has profited from advancements in well-structured settings, it is still reliant on a lot of annotated training data as well as hindered by the lack of uniformity in documenting in institutions of medical care.

Deep learning offers a potential solution through the use of neural networks to find patterns and connections within the huge unstructured information kept in EHRs. Unlike traditional methods, DL algorithms enable the identification of useful information in raw medical records without relying on detailed feature engineering. With the advent of transformer-based networks in deep learning (BERT, BioBERT, and ClinicalBERT), the phenomenon has allowed for revolutionary performance in critical NLP functions, such as named entity recognition. However, the application of such models to clinical text processing still presents challenges, such as problems of domain adaptation, the handling of unequal distributions of data, and enabling interpretability of the produced information.

To address these problems, the main objective of this study is the creation of an automated deep learning-aided means of extracting structured insights from unstructured electronic health records. The proposed method should minimize manual efforts in clinical information extraction and increase efficiency and flexibility. With domain-specific pre-training, multi-task learning, and entity disambiguation approaches, the system will enhance the identification and retrieval of critical medical concepts, relationships, and events. The results of this research will share a powerful and effective tool that can enable fast clinical decision support, medical investigations, and healthcare analytics, thus promoting the outcomes and efficiency of healthcare delivery.

1.4. Aims and Objectives

The purpose of this research is to create a deep learning framework that will support the automatic extraction of structured information from unstructured records in EHRs. The proposed method is designed to limit automated extraction of clinical data while improving accuracy, efficiency, and increasing the scalability of the system. By using cutting-edge deep learning technologies, this work will provide a higher degree of precise and timely identification of important medical details, interactions, and

occurrences in clinical notes, thus supporting clinical judgment and research, and healthcare data analysis as a result.

The aims and objectives of this research work are to:

1. To find out the deficiencies of the existing approaches
2. To develop an automated approach for structured information extraction from EHRs by overcoming the deficiencies of existing approaches.
3. To justify the proposed approach in terms of accuracy and efficiency using different case studies.

1.5. Research Questions

The following are the research questions of this study:

RQ1:

How can we automatically extract clinical information from electronic health records using deep learning?

RQ2:

How the proposed approach can outperform the existing methods in terms of accuracy, precision, and recall?

1.6. Research Contributions

This research brings major improvements in extracting the information from unstructured EHRs using Deep Learning methods. Unlike traditional rule-based and machine learning methods, the new approach uses a combination of deep learning and knowledge-based reasoning to improve the precision and efficiency of NER, RE, and the achievement of Information Structuring (IS) in clinical records. This work suggests a three-tier model for structured extraction, which enables a sophisticated medical entity, relationship, and contextual association categorization. More so, our work enhances the distinction of relationships and handles multiplicity issues chicly, overcoming major shortcomings in existing extraction models. Our methodology outperforms previous approaches in terms of precision, scalability, and flexibility, and its superior performance is reflected in comparison with state-of-the-art approaches. By automating extraction, this study improves clinical decisions, encourages medical

knowledge discovery, and hastens the advancement of artificial intelligence in healthcare applications.

1.6.1 Overview of Key Contributions

The chance to obtain structured data of unstructured EHR data is important in improving patient care, expanding medical research, as well as offering healthcare analytics. The majority of the current approaches include rule-based, machine learning, and NLP-based systems, which often have the problem of maintaining high accuracy, scalability, and flexibility. The accessible methods often do not find interrelationships between clinical entities and comprehend significant background, including dosage of medications, duration of treatment, and patient history.

To overcome these problems, a deep learning-based approach and a knowledge-based reasoning are suggested in this research. With the introduction of deep learning models to the knowledge of medicine, it is possible to extract the information that will be more precise by the system. In addition, the proposed approach uses an alternative three-level architecture that puts clinical entities, as well as their relations and dependencies, in an effective framework. Unlike the traditional approaches, the proposed framework is empowered with a higher level of comprehensiveness, scalability, and efficiency.

1.6.2 Integration of Deep Learning and Knowledge-Based Systems

The combination of deep learning and the use of knowledge-based systems is one of the major advancements in the given research related to medical information retrieval in unwritten EHR texts. The restrictions inherent in such conventional systems are that they are constrained by rules, which are not highly adaptable and confined in scale, but deep learning models can dynamically detect and leverage patterns in a huge range of datasets with no pre-direction. Nevertheless, on the side of the quite small details of the medical data (abbreviations, rare diseases, concealed relationships), deep learning methods can fail.

When employing domain-specific knowledge, this research advances the clarity and accuracy of the entities and relationships stated. The medical dataset is fed into a deep learning model, which identifies diseases, symptoms, medications, and their associations, after which the knowledge-based system validates and fine-tunes the

findings using medical standards. This combined approach improves the accuracy of NER and RE.

1.6.3 Precise Relationship Identification and Multiplicity

Its significant contribution to the literature is the exact extraction of complicated relationships and the successful manipulation of multiplicity among units of medical identity. The traditional approaches often ignore the intricate interrelations between diseases, symptoms, treatments, and drugs that may lead to incorrect or useless clinical conclusions.

This problem is addressed in this study through the design of a deep learning based relation extraction system, trained on large annotated collections of EHR information. Integration of the model can help researchers to better identify such associations, such as establishing relationships between drugs and diseases, between symptoms and diagnoses, or between treatments and their results. Moreover, the multiplicity that the model can manage enables it to determine dynamic interactions between several entities in one case, e.g., different medications for a single condition. These enhancements enable more reliable, organized expression of clinical data, facilitating better decision-making, risk prediction, and healthcare analytics projects.

1.6.4 Three-Tier Architecture Approach

The contribution of the research is mainly the design of a three level architecture that pays emphasis on the organized medical information extraction. Unlike the existing methods of strategy which often consider only the entity and relation extraction, this paper introduces a layered model to enhance the comprehension of the circumstances and the system as a whole usage.

Presentation Tier: Enables intuitive UI of clinicians and researchers to simply enter and analyze extracted data and present the data. It increases user involvement as it allows instant navigation throughout the discovered entities and their correspondences.

Logic Tier: Uses transformer-based architectures, such as BioBERT and ClinicalBERT, for NER and RE. It integrates a knowledge-based system to improve the precision and reliability of model predictions in the medical domain. Implementation of the logic layer in this research utilizes Python while using packages such as Natural Language Toolkit (NLTK), TensorFlow, and Keras.

Data Tier: Keeps and organizes extracted structured data to facilitate the latter stages of analysis, retrieval, and health care system integration.

1.6.5 Comparison with Existing Work

Rule-based systems, standard NLP techniques, or the application of ML algorithms are the most of assurance of past work in medical information extraction. However, standard methodologies often struggle to adapt to new domains, identifying links, large datasets in practice. This research differentiates itself by:

The integration of deep learning and knowledge-based system allows to make clinical information more accurate and interpretable. This mixed methodology will overcome major limitations in the identification of relationships and it can practically deal with the multiplicity and complexity of large clinical data-sets which are major limitations of the conventional approaches. The system is applicable to the rapid healthcare setting by not only extrapolating the relevant clinical concepts in real-time, but also dynamically analyzing the relationship between them, which is why it is helpful in quick-paced healthcare settings. Also, a three-tier architecture design is beneficial in terms of system usability, enhancing data storage and increasing the processing performance. This leads to a more effective and scalable clinical information extraction solution and eventually leads to improved decision support and more accurate downstream solutions like automated class diagram generation.

In the present case, this study gives a comparison of the proposed method with the state-of-the-art models, and the proposed method is also better in terms of precision, recall, F1-score, and scalability, which proves its effectiveness in deriving clinical information.

1.6.6 Impact and Implications

Today, the given research offers new guidelines to the field of medical AI to comprehend and enhance it. The information extraction system is to be proposed in a deep-learning-based system that has a few significant benefits:

Enhanced Clinical Decision-Making: The creation of a patient information structure will help to provide access to the medical team, therefore, allowing the health care specialists to deliver a more informed and stronger diagnosis.

Improved Medical Research and Analytics: EHR data is organized in a way that enables scientists to reveal the patterns, trends, and associations, which could not be identified previously because of the unstructured clinical records.

Reduced Manual Effort: The automation of the process will make the process lean, allowing the clinicians to spend more time caring for the patient than performing some of the tasks of data extraction.

Scalability and Adaptability: This system has three-tier architecture; hence, it can be applied in medical artificial intelligence platforms, research centers, and hospitals.

This groundwork is what the research lays to create a path to future advances in medical NLP and deep learning, creating a flexible and efficient way of arranging clinical data.

1.7. Structure of the Thesis

The figure below outlines our plan of work to conduct this research by providing an overview of the thesis structure, outlining the flow and organization of the chapters.

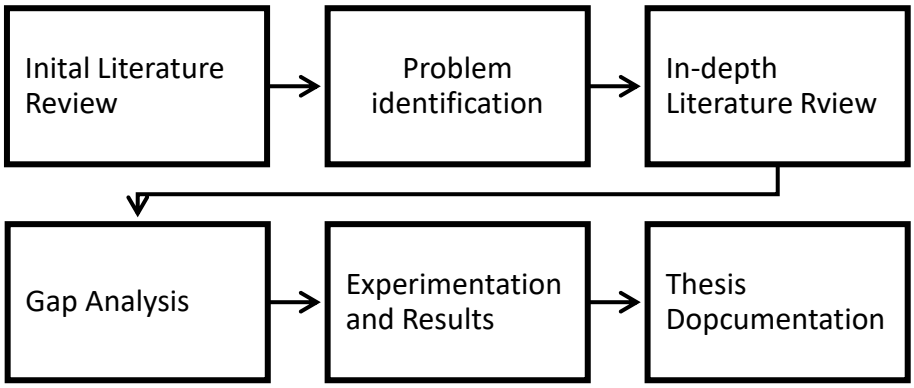

Figure 2. Thesis Structure

In chapter two, the most recent developments in using deep learning approaches to extract information from EHRs are discussed. The chapter begins by outlining the basics of the NLPs for IE, highlights the need for the extraction of structured data from clinical documents, and mentions the hardships encountered during this process. A discussion of the traditional methods (such as rule-based rule sets and machine learning approaches) that encompass the challenge that these traditional methods pose in processing the complexity and variation in medical language. It also explores how contemporary deep learning methods have assisted with better comprehending and

processing medical literature, such as transformer models, such as BERT, recurrent neural networks, and convolutional networks. By way of contrast and comparison of these approaches, the chapter exposes how the recent advances in IE from EHRs have greatly enhanced the extraction of medical information in terms of precision and speed.

In chapter three, we discuss our research methodology and how deep learning methods were employed to obtain structured data from electronic health records. In the beginning, the chapter describes why deep learning has been used instead of classic learning and presents used deep learning models with a particular emphasis on transformers and recurrent neural networks. Detailed information about data preprocessing stages is provided; the fitting methods of clinical data before training the model are described. The data is chemically processed through tokenization, entity identification, etc., to conform the information to the format for the input that is expected from the models. The chapter concludes with the experimental setup, for example, providing a description of the amount of computing resources that is needed, the deep learning tools used, and the metrics for evaluating the effectiveness of a model.

In chapter four, the experimental results and performance of the proposed deep learning-based model for Information Extraction from EHRs are presented. The chapter compares the proposed method with other state-of-the-art techniques in terms of accuracy, precision, recall, and F1-score, providing both qualitative and quantitative evaluations. The results are further enhanced by visualizations which include precision-recall curves. It helps in performance illustration across various metrics. Moreover, the chapter also explains the challenges faced during research. It help in identification on limitations of the research i.e. the need for large annotated datasets and the interpretability of deep learning models in clinical settings.

In chapter five, the detailed discussion of the experiment and evaluation is given. It gives the comparison of our approach from the existing approaches to acknowledge the strengths and weaknesses of the proposed approach. The discussion highlights the implications of the research to improve the decision-making and patient care. It also discusses the limitations of the research and their possible solutions, such as data quality issues and challenges in domain-specific terminology. Furthermore, it outlines directions for future research, including exploring unsupervised learning techniques and improving the explainability of deep learning models in healthcare applications.

The final chapter concludes the thesis with the important findings and contributions of the research. It addresses the answers of the research questions formulated and showed that the proposed research is improved in terms of accuracy and efficiency due to the use of deep-learning along with rule-based knowledge base for run-time entity prediction. The chapter also discusses the limitations encountered during the study and suggests areas for future work, such as integrating the model with real-time clinical decision support systems and expanding its applicability to other healthcare domains. It concludes with a reflection on the broader impact of this research, highlighting the potential of deep learning to revolutionize the extraction and utilization of medical data, ultimately improving patient outcomes and streamlining healthcare processes.

Chapter 2

Literature Review

2. Literature Review

It is necessary to go over the current approaches to automatically extract clinical information from EHRs before proposing our own method. The literature review will point out key aspects of current methods, and this will introduce the need for our research.

2.1. Overview of existing techniques

IE from EHRs has grown quickly lately, supported by the use of advanced NLP and deep learning techniques. Several key studies have suggested new approaches to enhance clinical information extraction accuracy, covering use cases from disease detection to risk assessment. This section discusses recent developments and focuses on important techniques, models, and issues in this field.

2.1.1. Question-Answering-Based Pipelines for Information Extraction

According to (Ying et al., 2022), they propose a novel Question Answering (QA) pipeline to extract data from EHRs. The approach solves the problem of scaling training data by automatically generating QA pairs from annotated EHRs. The method starts with cleaning up the text and analyzing its dependency tree, and then makes use of a QA model to answer different clinical questions. Finally, the model is refined to increase its practical use in real clinical situations. Even though it is a new approach, the method faced challenges due to the wide range and complexity of the questions handled in extraction. For example, questions about diagnoses and disease classification proved to be the hardest because of the complex language and inadequate training data..

2.1.2. Deep Learning for Relation Extraction in Radiology Reports

(Jain et al., 2021) worked on the RadGraph Benchmark, which is a deep learning model for finding both entities and relationships inside chest X-ray reports. They worked with the RadGraph dataset, which is made up of structured annotations for both clinical findings and their relationships. There were some issues cited, even though the model did well in relation extraction. The reports included in the dataset were mainly from a limited number of U.S. chest X-ray hospitals, making it hard to say if the model will work for other places and kinds of reports. Also, variations in how findings were

annotated, along with having fewer CheXpert patients from insufficient groups, increased challenges for the model's performance.

2.1.3. Pretraining BERT Models for Radiology QA Tasks

(Dada et al., 2024) examined how to use BERT-based models to extract information out of poorly structured radiology reports that contain natural language queries. Training on roughly 850,000 unlabeled reports and later using a medical encyclopedia were steps in the approach to improve the F1 score for answering questions. The study achieved encouraging results in performance; however, the model was not very effective in cases involving multiple or opposing pieces of information. In addition, the model's reasoning could not be easily interpreted, which could stop it from being adopted in clinical environments. The research showed that transparency and explainability are key factors for the use of QA models in healthcare.

2.1.4. Zero- and One-Shot Learning Using LLaMA-2 for Clinical Feature Extraction

The study by (Isabella C. Wiest et al., 2023)] focused on using LLaMA-2 to find clinical features linked to decompensated liver cirrhosis in MIMIC-IV records. Their method was based on approaches that involved zero- or one-shot learning and did not use any model fine-tuning. Prompt engineering methods and different model sizes were used, and the predictions were compared to a gold standard set by three medical experts. Although it was good at finding explicit features, the method was not as effective with features that were only in the text. Also, the study revealed that LLMs have their own biases, such as inferring patient gender from statistical prevalence rather than textual identifiers, which poses significant ethical concerns in clinical settings.

2.1.5. Evaluating LLMs in Adult Critical Care Contexts

According to (Lio et al., 2024) general-purpose Large Language Models (LLMs) like LLaMA-2-7B and GPT-4 were tested for their abilities in working with critical care reports. Concepts from critical care reports were extracted using MetaMap 2020a, and the team of clinicians used the CADA tool to annotate them. The models were given the tasks of spotting meaningful concepts, evaluating how relevant each concept was, and dealing with negated ideas. The models were fine-tuned with LoRA, and 300 GPT-4 responses were reviewed qualitatively. This study is majorly praised for its layered method of assessing clinical reasoning. Still, since all data were from one

center, the results may not be generalizable. On top of that, turning to third-party LLMs for this task introduces problems related to data safety and institutional policy mandates, which have not yet been addressed.

2.1.6. Deep Learning for Classification of Social Determinants of Health

(Han et al., 2022) worked on extracting SDOH from free-text EHR data by applying deep learning-based NLP models. Experts guided the building of an SDOH taxonomy, and three models were evaluated by the study. Researchers used a CNN, a BiLSTM, and a BERT classifier. The database consisted of 3,000 clinical records that were labeled by humans. BERT was better than standard machine learning mechanisms in terms of both accuracy and recall, according to the results. Still, since it was only conducted in one healthcare system, it is not clear whether the results can be trusted elsewhere. In addition, the biases that arose during manual annotation can affect how well the model learns and how it is used in later applications.

2.1.7. Memory-Based Neural Networks for Mortality Prediction

(J. Song et al., 2021) made use of a local-global memory network to record the development of individual patients alongside population-wide patterns for mortality prediction purposes. The model uses LSTM-based recurrent layers and uses data from four intensive care units in Denmark for training. Patient-specific information was recorded by the local memory, and the global memory picked up on general disease patterns. Even though the model could make accurate predictions, its lack of interpretability makes it hard for doctors to trust and use it.

2.1.8. Predicting Bacteremia Using Time-Series EHR Data

(Park et al., 2020) used Gated Recurrent Units (GRUs), a kind of recurrent neural network, to predict bacteremia in surgical patients by analyzing time-stamped records from 36,023 medical records at a tertiary hospital in South Korea. It observed patient records from ICU and general wards all the time to try to identify bloodstream infections as soon as they began. It was accurate, yet the model was not interpretable since the inner workings of the RNN were impossible to understand. Clinicians were unable to check or interpret the prediction of the model, making it less likely the model could be used in real hospital environments.

2.1.9. Time-Aware Attention Networks for Risk Prediction

HiTANet, the Hierarchical Time-Aware Attention Network, was introduced by (Luo et al., 2020) to help with predicting clinical risks from EHR data. This model made use of embeddings that considered time to model how patient information changed over time. When tested on three actual health records datasets, the model achieved solid F1 metrics and made risk prediction for diseases scalable. Even with its merits, the method had problems with data irregularity that often occurs in EHRs and remained difficult to interpret, which made it difficult for clinicians to evaluate.

In these studies, several trends and challenges become clear. Both transformer-based architectures and recurrent neural networks have achieved good results in specialized tasks, such as question answering, risk prediction, and relation extraction. In general, most approaches have problems with how their results can be explained, as well as with handling biased and not easily transferable data. Trained models usually struggle to apply to different contexts if there are changes in how things are documented, the population, or the data quality.

Researchers are increasingly using large language models and techniques like zero-shot and few-shot learning, which are promising for working with less training data. Still, the issue of mistaken facts, biases, and the inability to explain decisions needs to be resolved for these models to be used clinically.

In conclusion, while significant progress has been made in information extraction from EHRs using deep learning and NLP, future research must focus on enhancing model interpretability, addressing data heterogeneity, and ensuring fairness and ethical compliance in clinical AI systems.

Table 2

Comparative Analysis of Existing Literature

Paper	Methodology	Strengths	Gaps
(Ying et al., 2022)	Extracting information from EHRs using QA techniques, leveraging annotated data for training QA models, and	Excellent performance in information extraction, capable of handling few-shot or zero-shot settings, emphasizes the	1- Addressing diverse question types and constrained answer options in EHR extraction.

	implementing a pipeline with preprocessing steps.	importance of transfer learning and each component in the proposed pipeline.	2- Improving few-shot or zero-shot capabilities for handling scarce or misaligned annotated data in medical concepts.
(Jain et al., 2021)	Information extraction schema for entities and relationships from radiology reports using the development of RedGrapgh.	1- High coverage of clinically relevant information in reports. 2- Achieved a micro F1 of 0.82 for relation extraction on the MIMIC-CXR test set	1- The schema used does not capture clinical context in radiology reports beyond specific sections. 2- Cases of ambiguity in reports. 3- The dataset is limited to chest X-ray radiology reports from specific datasets and hospitals. 4- There is some inter-observer variability in the test dataset annotations. 5- The demographic attributes in the datasets are not consistently balanced
(Dada et al., 2024)	The methodology is evaluating BERT models on unlabeled radiology reports and	1- High F1 score of 83.97% and an exact match score of 71.63% for answerable questions.	1- Restriction of answers to a single, consecutive text span.

	annotated reading comprehension datasets, with continued pre-training on radiology datasets and a medical encyclopedia resulting in the most accurate model.	2- 96.01% accuracy when detecting unanswerable questions. 3- Efficient way to extract information from radiology reports, reducing manual searches. 4- Good generalization over unknown question-answer pairs.	2- The challenge of comparing reports from various radiologists due to different styles and word choices. 3- Overestimation of model scores in determining the position of an answer.
(Isabella C. Wiest et al., 2023)	Developed an open-source pipeline using a locally deployable Large Language Model (LLM) called “Llama 2” for extracting quantitative information from clinical text.	1- Demonstrated effectiveness in identifying key clinical features like liver cirrhosis, ascites, abdominal pain, shortness of breath, and confusion from patient medical histories. 2- Lightweight LLM “Llama 2” showed excellent performance in extracting clinical data, achieving high accuracy without task-specific retraining. 3- Rigorous, quantitative analysis comparing model outputs to ground truth obtained by medical experts.	1- Limited grasp of contextual subtleties for extracting implicitly mentioned features like confusion. 2- LLMs based decisions on probabilities rather than clear identifiers for certain tasks, leading to potential biases.

(Lio et al., 2024)	MetaMap and clinician annotations were applied on MIMIC-III nursing notes, evaluated by LLMs and fine-tuned models, with performance assessed through qualitative review and statistical analysis.	1- A comprehensive approach was used to assess the proficiency of LLMs in processing real-world clinical notes from adult critical care settings. 2- Varied assessments for detection, encounter, and negation tasks. 3- The curated dataset of clinical concepts and notes serves as a resource for future research in the field of LLM evaluation in healthcare.	1- Using single-center data may limit generalizability to other centers. 2-Testing LLMs external to an institution presents potential barriers to ensuring data security and adherence to institutional policies. 3- The study's findings' long-term applicability may be limited by the rapidly evolving nature of LLM technology and clinical practices.
(Han et al., 2022)	Developing a framework to create a set of SDOH categories based on existing taxonomies and expert input then Evaluating three deep learning models: CNN, BiLSTM, and BERT-based Trained models on a manually annotated dataset of over 3,000 clinical notes.	1- Demonstrated improved performance of deep learning models over traditional machine learning baselines in classifying SDOH mentions BERT-based model achieved the best overall performance.	1- Limited to a single healthcare system. 2- Validation across different clinical settings is needed to assess the generalizability of the approach. 3- The manual annotation process used to create the training data could introduce biases that

			impact model performance.
(J. Song et al., 2021)	Utilizes local-global memory neural networks (based on LSTM) for mortality prediction by learning both individual patient patterns (local memory) and global disease evidence (global memory) from ICU data.	Captures both personalized and generalized information for better predictive performance; designed for longitudinal patient data.	1- Lack of model interpretability (black-box nature). 2- Limited insight into decision rationale reduces clinical trust.
(Park et al., 2020)	Applies RNNs (specifically GRUs) on time-series EHR data from over 36,000 surgical inpatients to detect bacteremia; data sourced from ICU and general wards.	Real-time, continuous monitoring; effective for infection prediction in dynamic clinical environments.	1- Model opacity makes it difficult to understand prediction logic. 2- Limited to data from a single hospital, affecting generalizability.
(Luo et al., 2020)	Implements a transformer-based hierarchical attention network incorporating temporal information from EHRs to enhance risk prediction accuracy across three real-world datasets.	Captures temporal dependencies; performs well on irregular clinical data; scalable for risk prediction tasks.	1- Struggles with irregularities in EHR documentation. 2- Lack of model explainability; decision-making process remains opaque.

Chapter 3

Experimental Work

3. Experimental Work

3.1. Proposed Research Framework

We implemented a BiLSTM NER pipeline and tested it on five different datasets to see how well it generalizes and how durable it is. All our datasets had medical text annotated with brackets that, for example, mean PROBLEM is a named entity.

Extracting the data from each dataset was the very first task, performed by the load_data function. Sentences were processed so that each dataset's raw text is turned into sequences of tokens and labels. Each dataset is preprocessed in a way that empty lines or punctuation at the end separate out the sentences for proper segmentation. Data in every dataset was cleaned so that the unique parts are set as tokens with the proper labels.

Following data extraction, prepare_data was used to change the sequences into the right format for training the model. It built two dictionaries: Using word2idx, each different word was assigned a unique label, and tag2idx did the same for entity tags. Sequences were made all the same length by using Keras' pad_sequences utility. Labels were transformed into vectors for each option in the class using to_categorical, to be used with multi-class token-level classification.

For each data set, the model was checked on fresh data by splitting it into an 80/20 training-testing section using train_test_split. Building all the models in the same way was possible because they were all built with Keras' Sequential API. An Embedding layer was used first to map each token to a dense vector, and afterward, the Bidirectional LSTM layer looked at the token's earlier and later words. Because the Time Distributed Dense layer was activated with softmax, the model gave each word a probability distribution over the available entity classes.

Adam was chosen as the optimizer, and categorical cross-entropy was used for loss because the model performed multi-class classification. Training was conducted over 20 epochs with a batch size of 32 and a 10% validation split from the training data to monitor overfitting and performance during training.

To enhance the real-time understanding of medical details in clinical text, a system was built that follows a set of standardized guidelines. It allows us to find out which

data points cause the model trouble and which it can deal with without errors, giving us ideas for improvement or adjusting the system for specific domains.

To help recognize and classify medical terminology in real time in clinical text, a medical knowledge engine was developed using rules. The technology relies on a specially-built dictionary to associate medical terms—for example, diseases, symptoms, conditions, and test markers—with their important attributes. All entries in the dictionary are under Disease, Symptom, or Test Marker and give lists of symptoms, tests used to diagnose, treatments, and conditions or diseases with similar symptoms. medical_rule_engine(word) was developed to use the data from the knowledge base. When a user types a disease, the system displays its symptoms, tests needed for diagnosing, and different ways to treat it; if a symptom is entered, the system shows the related diseases and the related diagnostic tests. Entering a disease generates a list of its symptoms, types of tests to recommend, and various forms of treatment. If a person reports a symptom, the result provides details on possible illnesses and suggests diagnostic tests to perform. Should the term not be included, the function reports that no data is found. By using this method, individuals can offer input related to specific issues and gain a deeper understanding of medical information found in studies and health care.

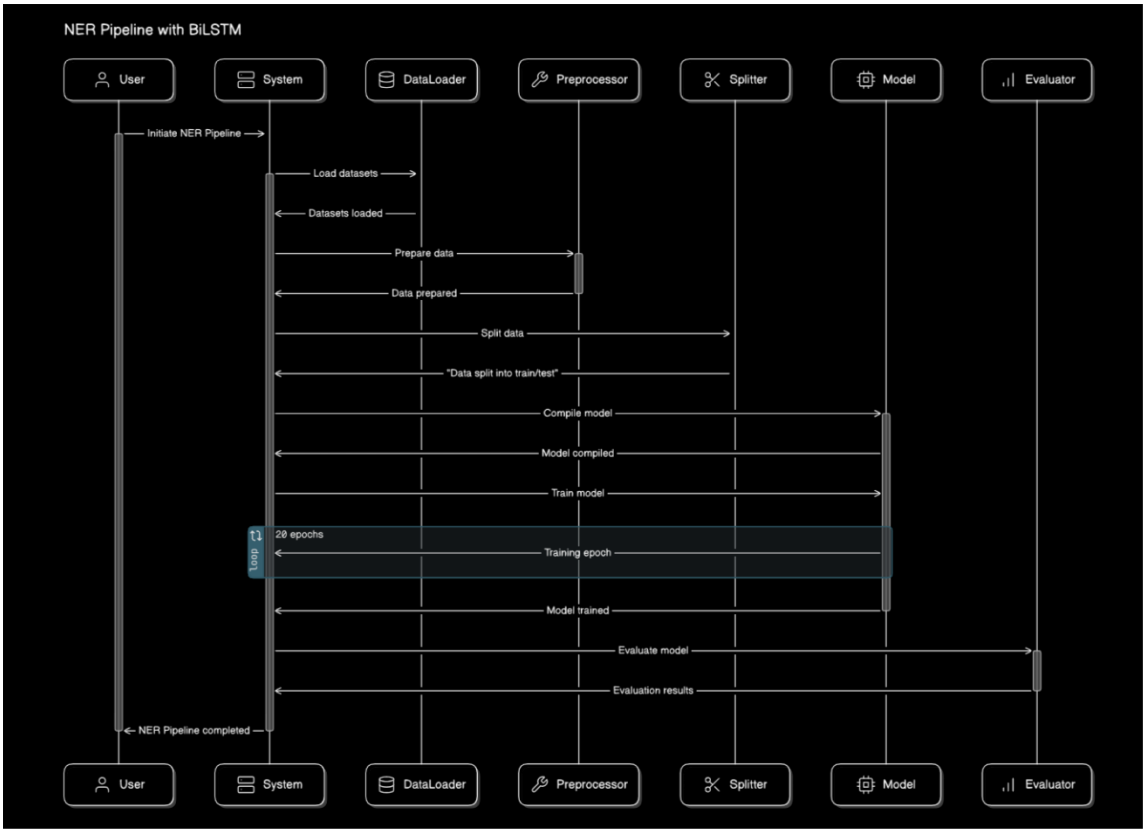

Figure 3.1. High-Level Implementation Framework

3.2. NLP Techniques

Through NLP, important information can be taken out of unorganized EHRs. Combining linguistics, computer science, and AI is what allows NLP methods to make sense of and analyze clinical documents. Before computational analysis of EHR data, it is necessary to use tokenization and POS tagging, as they are essential steps in NLP.

They work well in healthcare, as many important pieces of patient information are available in clinical narratives, such as doctor descriptions and lab results. With NLP, we can extract medical entities (for example, diagnoses, medications, procedures), their connections, and use this information to improve decision-making and help data systems communicate.

3.2.1 Tokenization

Tokenization means taking medical text and separating it into units called tokens, which may be words, subwords, or even characters. This is needed to prepare unstructured notes in EHR records for machine learning algorithms.

For example, in the sentence:

"The patient was prescribed 500mg of amoxicillin twice daily for seven days."

Tokenization splits it into:

["The", "patient", "was", "prescribed", "500mg", "of", "amoxicillin", "twice", "daily", "for", "seven", "days"]

When used in healthcare, tokenization must deal with medical terms, abbreviations, and numbers. When tokens are properly arranged, NLP tasks such as recognizing entities work better with medical terms.

3.2.2 POS Tagging

Each token in clinical text is given a grammatical class by POS tagging, which helps to analyze its structure. It helps you recognize the difference between a medical term, a word that shows an action, and a descriptor.

3.3. Model Selection

The correct model helps ensure the correctness of information taken from EHRs. Because clinical language uses a lot of distinct words and abbreviations and relies on the situation, strong techniques must be applied. It was found that deep learning was the best approach since it could identify patterns in unorganized text more efficiently. It outshines other machine learning models in activities such as NER and relation extraction, which is why it is perfect for this study.

3.3.1 Deep Learning Networks

It was decided to use deep learning models since they are able to automatically create hierarchical and abstract representations from a great deal of unorganized data. Deep learning uses neural networks with several steps, whereas conventional machine learning relies on the creation of hand-made features from data. Because of how words relate to each other in context, it is more effective in handling NLP.

Using deep learning helps to quickly and effectively find entities, attributes, and relationships from clinical narratives. While rule-based and traditional template approaches are helpful for controlled cases, they do not always perform well with the variation and uncertainty found in language. In addition, people must work hard to set up rules that may not work well with new input data. Instead, deep learning models get

insights from data, allowing them to become more flexible and general in solving problems.

Through the use of deep learning in this case, insights from clinical texts are automatically discovered and used to extract clinical entities. This allows for greater accuracy and a reduction in work that has to be done by hand, making the entire process more dependable and efficient.

3.3.2 Neural Networks

All the deep learning models in this study rely on the architecture of neural networks. Artificial networks link layers of artificial neurons that receive input, adjust it using weights, and activate them with functions. Training involves using backpropagation, which fixes the network's weights by considering the errors of the predictions, allowing it to learn even the most complex relations.

3.4. Model Architecture

How the model is built affects how effectively it is able to extract and use information for different problems. Relying on neural networks, the next step is to improve factors such as layer design, how many layers, what kind of activation functions to use, and which optimization algorithms to apply. The purpose is to make sure the model can uncover relationships, traits, and the numerous instances of these between classes in medical reports.

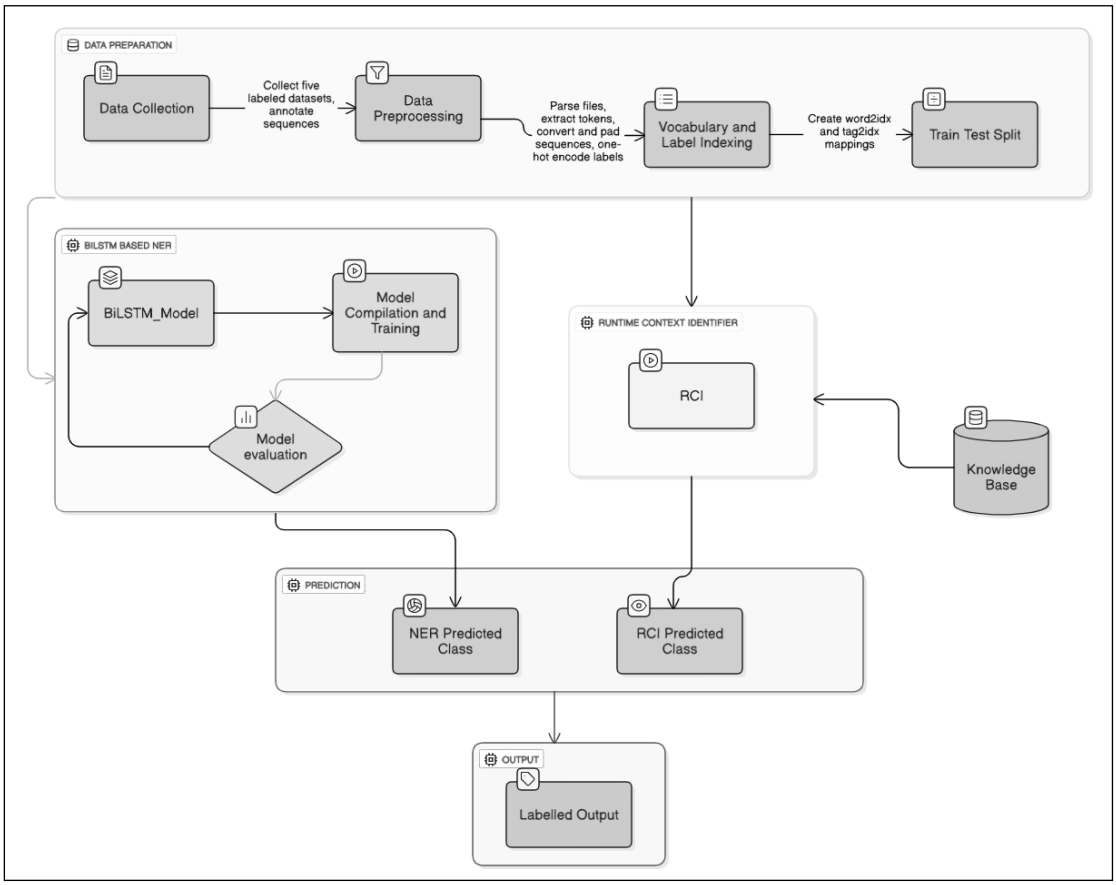

Figure 3.2. Research Methodology Architecture

Figure 3.2 shows the framework of the proposed method with details of all four phases. In the first phase of data preparation, five labeled datasets were annotated with relevant medical entities. These datasets underwent a preprocessing pipeline including the text preprocessing tasks like extraction of tokens, sequence padding, feature encoding and labeling, etc. The final step in this phase involved splitting the dataset into training and testing in an 80/20 ratio and preparing the data for model development. In next phase, we predict the class of the medical entity by using two different approaches that are BiLSTM-based NER and Runtime Context Identifier (RCI).During this phase, the BiLSTM model is trained using preprocessed data to predict the classes of different medical entities. The model is compiled and trained to predict the class of an entity. A key feature integrated into this stage is the Runtime Context Identifier (RCI), a rule-based medical knowledge engine designed to complement the NER model by providing real-time, context-aware predictions of medical entities. It is a dynamic engine to deal with the runtime entities, and is capable of taking new rules at runtime and then extracting the information related to the medical entity. The RCI system operates independently of the training limitations of the NER model. It utilizes a

comprehensive, custom-built dictionary that maps a wide range of medical terms, including diseases, symptoms, diagnostic tests, and treatments, to their associated clinical attributes.

Finally, in the Prediction phase, the trained model is used for live medical entity prediction. Users can input text in real time, and the model, guided by the RCI, returns relevant medical entities. The predictions are visualized through result dashboards, which include charts and performance plots for user interpretation and analysis.

3.4.1 Keras

Researchers in this project use the Keras API to easily design and evaluate neural network models. With this approach, it is easy to build complex neural networks very quickly. The complexity behind the scenes is handled by Keras, so that it is simple for researchers to define layers and other settings.

3.4.2 Tensor flow

In this work, TensorFlow is used to reduce the load on computers when training deep neural networks with EHR data. Thanks to parallelism and GPU use, training times become shorter, and developers can work with more intricate models.

Furthermore, TensorFlow helps integrate ready-to-use models into actual projects. Thanks to tools for deployment, tracking, and scalability in its ecosystem, it serves well in practical applications. For clinical information extraction, TensorFlow ensures that the neural network can be trained efficiently and deployed effectively, enabling consistent and scalable extraction of information from clinical texts.

3.5 Knowledge-Based System

In Knowledge-Based Systems (KBS), AI relies on pre-programmed knowledge to make decisions and solve problems. They use a set of organized information combined with a system that can make decisions using rules from the field. When human expertise is key, as in clinical informatics, KBS can supplement the data-based approaches used in the domain.

In practice, a way of organizing knowledge is merged to help determine which labels are suitable for particular entities. KBS uses domain-specific regulations and object-oriented concepts to ensure that all steps extracted are up to standards.

When you include a KBS, the proposed approach extracts clinical entities that combine practical information from data with the important rules involved in clinical systems. As a result, the process becomes more reliable and stronger.

3.6. BiLSTM-Based Named Entity Recognition Pipeline

In this section, we present an implementation of a BiLSTM NER pipeline and test its generalizability using five different datasets. However, each dataset contains medical text bracketed with labels where entities like PROBLEM appear in the text explicitly. First, we load data by the load_data function, which will parse a raw text file to a sequence of tokens and labels. Punctuation or empty lines are used to perform sentence segmentation to separate sentences correctly. Then the extracted data is cleaned and tokenized, and each unique term is assigned the correct label.

After data extraction, the prepare_data function takes the token and label sequences and transforms them into a format model training expects. Then, we create two key dictionaries, word2idx and tag2idx, which map each unique word and entity to a unique index, respectively. All sequences are padded to ensure uniform input length for the neural network with the Keras pad_sequences utility. Furthermore, labels are one-hot encoded with to_categorical so that they can be used for multi-class token-level classification tasks.

Using an 80/20 split, we can evaluate the performance of each dataset on data that it has not seen. Firstly, I use Keras' Sequential API to build the model using an Embedding layer that converts tokens to dense vector representations. Then we have a Bidirectional LSTM (BiLSTM) layer which processes input sequences in the forward and the backward directions (i.e, it has recurrent connections going backward as well) to capture context not only from past tokens but also from future tokens. Finally, a TimeDistributed Dense layer with softmax activation outputs a probability distribution over entity classes for each token.

The model is trained using a categorical cross-entropy loss function for multi-class classification problems and the Adam optimizer. Training is done for 20 epochs with a batch size of 32 and held out 10% of the training data as a validation set to watch the performance and to check for overfitting. The same setup of the model as seen here is repeated for all datasets, which provides a fair comparison and evaluation of the generalizability of the model.

3.7. Rule-Based Medical Knowledge Engine

A rule-based medical knowledge engine was developed to support real-time understanding of information related to medical topics in clinical text. This is a standardized system that helps recognize difficult data points for the model as well as regions that the model can rely on. It provides valuable diagnostic feedback for improvements or adaptations to a particular domain.

An engine is employed that consists of a curated dictionary that binds medical terms such as diseases, symptoms, conditions and test markers, to their respective information attributes. Then, each term in the dictionary is classified as Disease, Symptom or Test Marker which is also connected with relevant symptoms, diagnostic tests, treatments, as well as other concepts of interest to medical professionals. The function medical_rule_engine(word) is an interactive function that queries the knowledge base. The engine would output the symptoms, recommended diagnostic tests, and available treatments for a disease after a user inputs it. However, if you enter a symptom then the system suggests what illness it could signify and what tests to carry out to asses.

In the event that the input term is not found in the knowledge base, the system communicates this fact to the user. This interactive ability helps deepen the context understanding and assists researchers and healthcare professionals to comment and analyze medical subjects efficiently. Overall, the system developed through this hybrid rule-based and data-driven approach offers better reliability and insights to medical entity recognition.

3.8. Implementation Tools

The implementation of the project used a wide range of tools and technologies, specially selected when each was best suited to dealing with some aspect of development, data processing, machine learning, and the system itself. Below, there is provided an extensive examination of the main instruments and their particular contributions to the project.

3.8.1 Development Environment

Google Colab:

For this project, Google Colab was chosen as the major development environment, since it is cloud-based and user-friendly with great support for machine learning

workflows. Deep learning models could be built, trained, and evaluated in Colab, which proved to be the best platform to work with throughout the whole development. The ability to quickly experiment and train with Python code in the cloud, with free access to GPUs, meant we could make rapid progress on our class diagram generation system that we couldn't have otherwise.

The main reason I chose Colab is that it supports integration with most of the major machine learning libraries like TensorFlow, Keras, NumPy, etc. Their role was crucial in the project as both of these libraries were essential for NLP tasks and making LSTM-based sequence models. Furthermore, the platform made possible the seamless execution of code bound in data loading, data preprocessing, padding sequences, building the neural architectures (such as bidirectional LSTMs with dropout layers), and training neural models with categorical output.

In particular, Colab features built-in support for training and constructing complex neural network architectures with TensorFlow and Keras. For example, using the high-level APIs of Keras, a bidirectional LSTM model was created, and the training process was done on Colab's notebook interface within a reasonable environment. Being able to visualize and debug the progress of training and visualize outputs interactively proved extremely useful in being able to quickly prototype and tune model performance to perfection.

Moreover, Colab enables real-time collaboration, which played a major role during instances where multiple contributors worked on various parts of the project together. By sharing notebooks, the team could easily review and test each other's work. Moreover, Colab's support for GitHub integration made version control and tracking of code changes straightforward, enhancing overall team productivity and consistency.

An important point is that Colab made it simple to manage data preprocessing in a single line, for example by reading from a text file, highlighting expressions with regular expressions, converting tokens and tags to integers, and filling up sequences. I managed to include and test all these processing steps in the notebook, helping me stay organized and focus on smaller tasks.

Also, Colab notebooks allow you to document your work well because you can write explanations in between your code using markdown. Because of this, developers could

easily understand and duplicate the recorded steps in their work, allowing both collaboration and further work on the project.

As a result, using Google Colab allowed for the successful use of machine learning techniques in generating class diagrams. I was able to complete the project because of Jupyter's free computing, ability to share files, and large library collection.

3.8.2 Programming Language

Python:

The reason Python was picked for this project is its easy-to-read code, straightforwardness, and wide collection of libraries, all key when doing deep learning and labeling sequences needed for generating class diagrams automatically. Because the code is easy to write, the team made sure it could be maintained and kept clean, which benefited them when working on Google Colab.

Being able to use Python for data preparation, modeling, and evaluation was crucial for the present research. We relied on NumPy libraries for number-based tasks, while using TensorFlow and Keras to easily create, teach, and examine the Bidirectional LSTM model. Writing only a small amount of code allowed me to quickly train various types of neural networks.

Members of the Python community were also important. With its detailed documentation and active members on the online discussion forums, I received quick assistance when either tracking down bugs in the pipeline or improving the performance of my model. Having access to this system was very useful when we had to create custom logic for inputs from wide-ranging sources and domains.

All in all, Python was a perfect pick for this research because of its extensive library system, easy-to-write code and the fact it works with Google Colab. It made it possible to develop a system that learns from expert-created datasets, resulting in the automation of information extraction.

3.8.3 User Interface Library

Google Colab was used as the primary user interface environment for this project because it has a simple interface, is interactive, and connects well to Python. Google Colab was chosen to host this project. Google Colab gives users a simple interface in

notebook form, which is more convenient for carrying out research in deep learning and natural language processing.

For this research, the Colab environment served both as the development and user interaction interface in the sense that the research team was able to upload EHR datasets, do some pre-processing, and link the output for deep learning model definition and training, and visualize the results within a single browser-based platform. Users could provide input (e.g., dataset file path) via interactive cells, view real-time feedback (e.g., model summary, evaluation metrics), and further change model configuration live at runtime.

In addition to the visualization of the classification results, performance metrics, and sample predictions, Google Colab also makes it easier for us to understand how the model is extracting clinical entities such as diagnoses, medications, symptoms, etc. The interactive format of the system proved to be a smooth workflow to debug, experiment, and refine the system as the model changed from handling structured medical data to handling unstructured medical data.

On the other hand, the notebook-style interface also served as documentation and demonstration: it was easy to follow the research process step by step. Rapid prototyping, reproducibility, and the ability to share experiments, all critical in machine learning research for sensitive and complex data such as EHRs, were enabled.

Overall, using Google Colab instead of a traditional Graphical User Interface (GUI) removed the need for a GUI since this open, robust, and rich environment made it most suitable as a systems user interface, able to support the entire research cycle, from data loading to model interpretation, for this deep learning based information extraction system.

3.8.4 Natural Language Processing Library

NLTK:

The main library used in this research to implement NLP approaches was NLTK. NLTK is a complete package that will allow you to write Python programs that interact with human linguistic data, and it is the best tool that you can use to process the textual clinical data needed to extract medical entities.

Tokenization is one of the most important NLP techniques with NLTK and is simply the process of dividing the text into isolated pieces called tokens, which are usually words or phrases. One of the first fundamental stages in the processing of the text is tokenization, since it makes it possible by converting unstructured text into a format suitable for further analysis. In this thesis, I used tokenization to find possible relations and medical entities within the text. Segmenting the text into discrete tokens can cause the algorithm to better assess and pull the necessary details for information extraction. Another critical NLP approach employed with NLTK is POS tagging, assigning each token with a part of speech, i.e., a noun, verb, or an adjective, etc. POS tagging gives the tokens syntactic context so that the system can understand how words are organized and what meaning they can convey.

The vast library and toolkit that NLTK provides made it easier to do some text analysis and manipulation for this project. NLTK was used in the study, where the provided textual input was processed, and the information collected was accurate and relevant to extract information from EHRs. Ultimately, the NLTK NLP approaches made for these natural language descriptions automated the information extraction from the natural language medical data.

NumPy:

Numerical Python (NumPy) is a powerful Python library that supports working on large multi-dimensional arrays and matrices as well as several useful mathematical operations on such enormous arrays. This research mostly uses NumPy to perform mathematical operations and process numerical data to train a model. Because it can process large amounts of data easily, there is no shortcut for prepping data to feed into machine learning models, and therefore, it is an important tool because it can pull off tasks such as statistical analysis, random number generation, and linear algebra. NumPy helps us manipulate arrays because the data needs to be organized in the format required by deep learning models and fed to those models in such a way that data input is at maximum speed and efficiency.

SpaCy:

SpaCy is a Python package for robust and high-performance Natural Language Processing (NLP). In this work, named entity recognition (NER), tokenization, dependency parsing, and part-of-speech (POS) tagging tasks of domain-specific NLP

tasks are conducted by using SpaCy. To extract precise information from Electronic Health Records (EHRs), the ability to accurately obtain relevant data from unstructured clinical narratives is essential. SpaCy provides a streamlined NLP processing pipeline and pre-trained models that are significantly faster and more scalable for handling large medical datasets compared to other NLP libraries. In this research, SpaCy is utilized to process complex clinical text and efficiently convert it into structured data. This capability enables the system to quickly identify medical entities such as diseases, symptoms, and test markers, as well as their relationships, facilitating accurate and real-time extraction of clinically relevant information.

3.8.5 Machine Learning Framework

Keras:

The deep learning models were created, implemented, and trained using a high-level neural network library called Keras. With an intuitive programming interface (API), Keras makes it easy to experiment quickly with a new neural network structure. Because the model is modular, layers, optimizers, and activation functions can be easily added or changed to adapt it to a different task. Keras helped in building the neural network models for this research, making associations and several classification models from this data. Keras gave us the flexibility to swiftly and effectively iterate and ensured that the models were appropriately suited to the challenge at hand.

The ease of Keras's intuitive API enabled the research team to quickly test out different network designs and configurations of components that were fit for purpose. Speedily prototyping and refining the models using Keras shortened the development process, leading to more precise and better models.

Tensor Flow:

Keras simply served as the frontend framework, where the model is built, inspected, and deployed, while Tensor Flow acted as the backend engine to provide the logic necessary to train and deploy the neural network model. Some of Tensor Flow’s features include support for distributed computation, automated differentiation, and advanced tensor operations, which are crucial for training a deep learning model in a large-scale dataset. TensorFlow's extensive API provided fine-grained control over every facet of the model's architecture and training procedure, and its scalability allowed the project to take advantage of many GPUs for speedier training times. The

comprehensive documentation and robust community support of Tensor Flow were particularly helpful tools for debugging and optimizing the models.

In this work, TensorFlow is used to reduce the load on computers when training deep neural networks with EHR data. Thanks to parallelism and GPU use, training times become shorter, and developers can work with more intricate models.

Furthermore, TensorFlow helps integrate ready-to-use models into actual projects. Thanks to tools for deployment, tracking, and scalability in its ecosystem, it serves well in practical applications. For clinical information extraction, TensorFlow ensures that the neural network can be trained efficiently and deployed effectively, enabling consistent and scalable extraction of information from clinical texts.

3.8.6 Data Processing and Integration Tools

The point I'm trying to make is that any machine learning project needs data processing and integration. However, working on big and complicated datasets, such as an Excel file, is no less important. After considerable care was taken to process the Excel data to ensure that it was in the best possible format for training the neural network model, the data was embedded into the machine learning workflow.

The first step was using the Python Pandas module to read and import the data from the Excel file, as it required some fitting according to the structure. This is just a great assignment to use pandas since it comes bundled with a robust and flexible set of tools to work with and analyze data. After the data was loaded in a Pandas Data Frame, several preprocessing tasks were performed, which included handling missing values, normalizing or standardizing the features, and encoding the categorical variables. To ensure that data is clean, consistent, and ready to feed into the neural network models, a couple of preprocessing steps are required.

Following preprocessing, the data was transformed into a format that Keras and TensorFlow could use, which was usually NumPy arrays. After processing, the data was divided into training and testing sets so that the models' performance on previously unseen data could be assessed and adjusted.

Chapter 4

Result and Discussion

4. Results and Discussion

All the results obtained must be properly explained with the help of figures, graphs, and tables.This would allow us to represent the results of the conducted research and experiments in this chapter.

This information is already covered in the earlier chapters on using Python as a programming language and the use of TensorFlow and sklearn as machine learning libraries. A Bi-directional Long Short Term Memory (BiLSTM) model and a NER pipeline were implemented, and the procedure was conducted on five distinct datasets to assess the model's generalizability and strengths.

4.1 Experimental Setup

In this experiment, a classification model is evaluated across 5 different data sets that are likely to have different data characteristics, for example, feature distribution, sparsity of samples, class unbalance, or domain specificity. Multiple datasets were used to ensure that the model was robust, adaptable to various types of input conditions, and could generalize well. Three fundamental metrics, accuracy, precision, and recall, were used to evaluate the model, but each gives an entirely different view of the performance of the model.

We used the confusion matrix to find these measures. The four main segments in which the confusion matrix is distributed are as follows:

- **True positives (TP):** The cases in which a positive outcome is determined to be appropriate are considered as True Positives.

- **False positives (FP):** The cases with negative outcome but labelled as good by mistake comes in the category of False Positives.

- **True Negatives (TN):** The cases with negative outcome and also were correctly labelled in the category of negative impacts.

- **False negatives (FN):** The cases with positive outcome but labelled in the category of negative impacts.

With the use of confusion matrix we may calculate multiple measures such as precision, Recall and F1 score.

- **F1-Score:** It is the harmonic mean of precision and recall and it balances both.

Formula:

$$\boldsymbol{F1 - Score = \frac{2(Precision\,.\,Recall)}{Precision + Recall}} \qquad \mathbf{1}$$

- **Precision:** The proportion of the cases that were labelled positive and have a positive outcome is referred to as "precision".

Formula:

$$\boldsymbol{Precision = \frac{TP}{TP + TN}} \qquad \mathbf{2}$$

- **Recall:** The ratio of all the TP to the total sum of TP and TN is known as "Recall".

Formula:

$$\boldsymbol{Recall = \frac{TP}{TP + TN}} \qquad \mathbf{3}$$

Table 4.1

Performance Evaluation Measures

Sr#	Metrics	Formula
1	F-Score	$F1 - Score = \frac{2(Precision\,.\,Recall)}{Precision + Recall}$
2	Precision	$Precision = \frac{TP}{TP+FP}$
3	Recall	$Recall = \frac{TP}{TP + FN}$

The experiments have shown the following results:

Table 4.2

Experimental Results and Accuracy Matrix

Dataset	F1-Score	Precision	Recall
1	0.91	0.92	0.91
2	0.91	0.91	0.91
3	0.94	0.94	0.94
4	0.88	0.89	0.88
5	0.95	0.95	0.95
Average(1-5)	0.921	0.932	0.921

The graph added below shows the results of the conducted experiment in a pictorial representation.

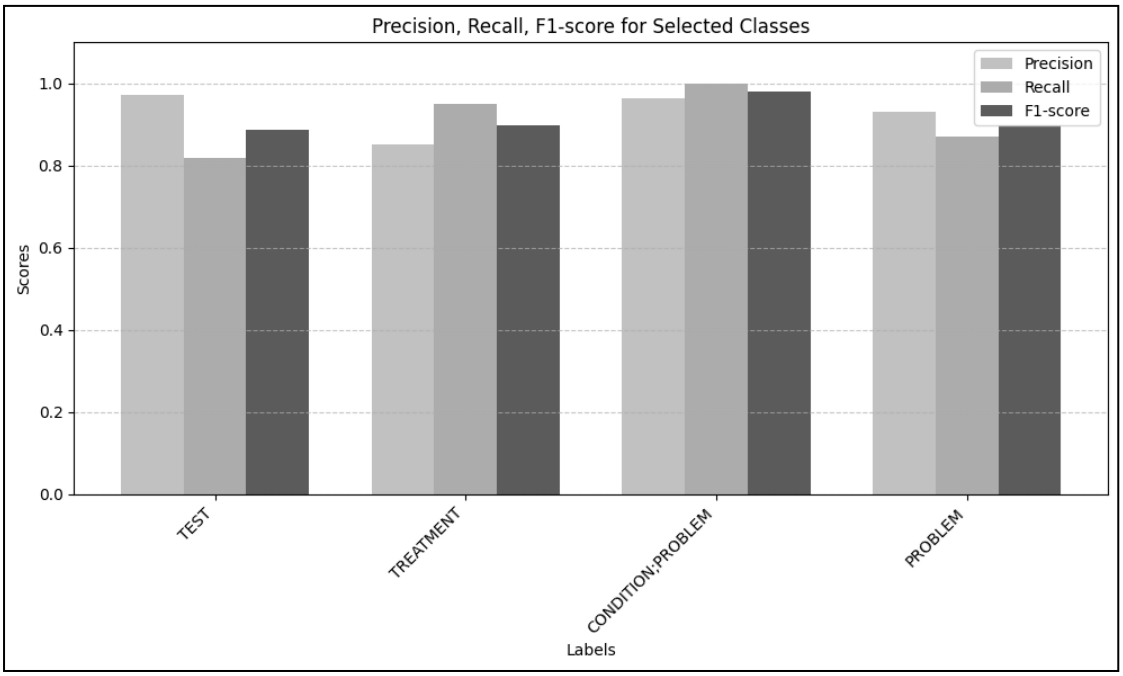

Figure 4.1. Pictorial Representation of Results of Dataset 1

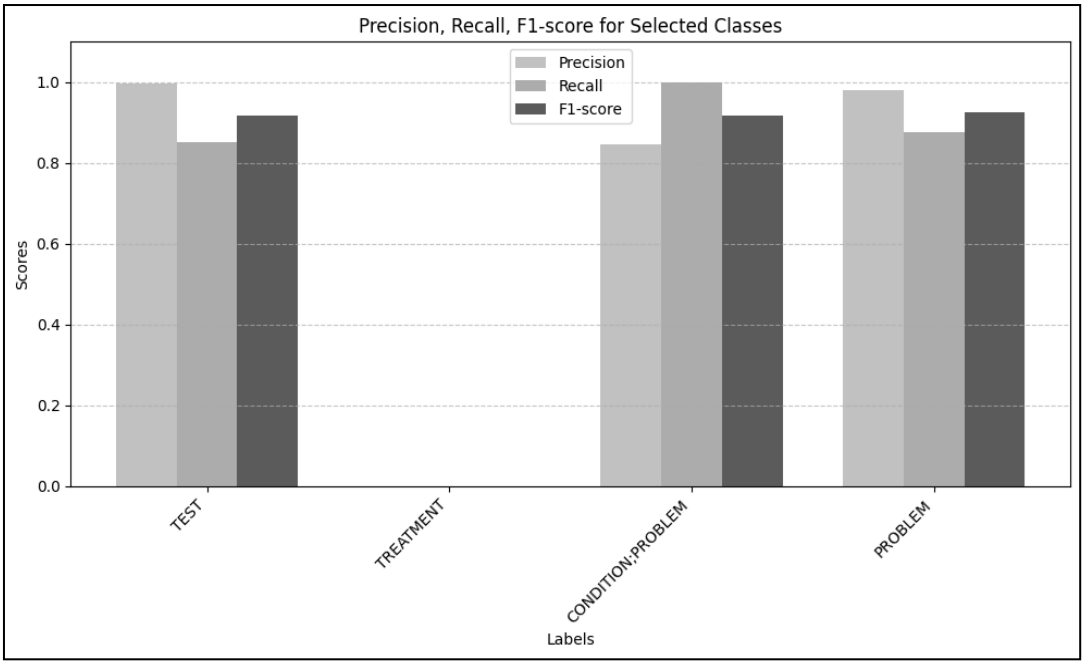

Figure 4.2. Pictorial Representation of Results of Dataset 2

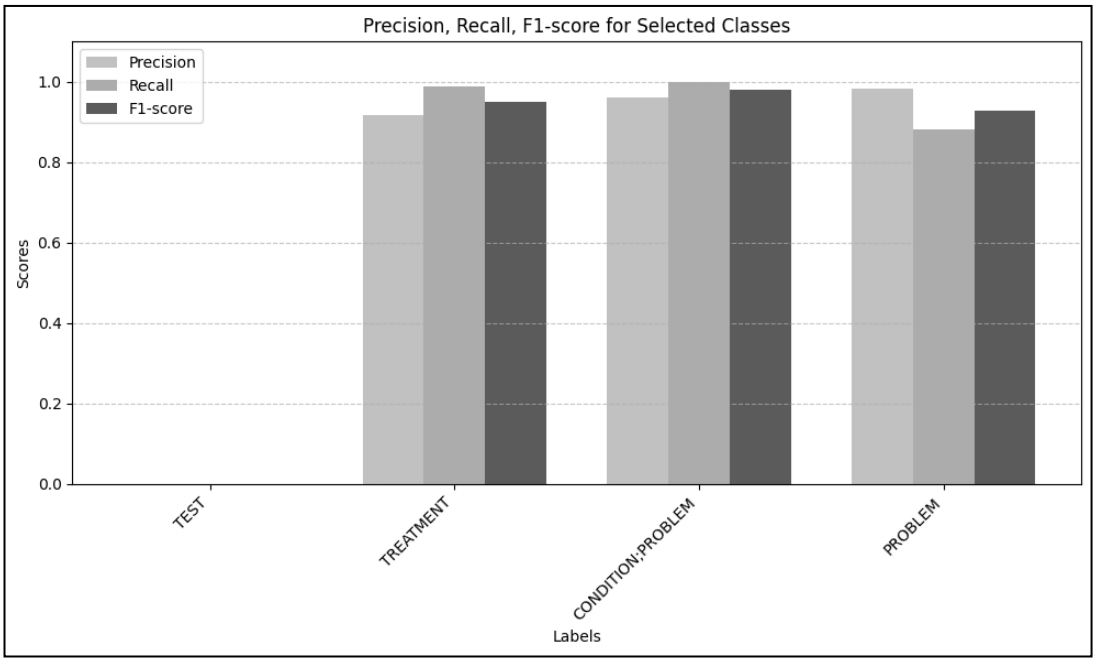

Figure 4.3. Pictorial Representation of Results of Dataset 3

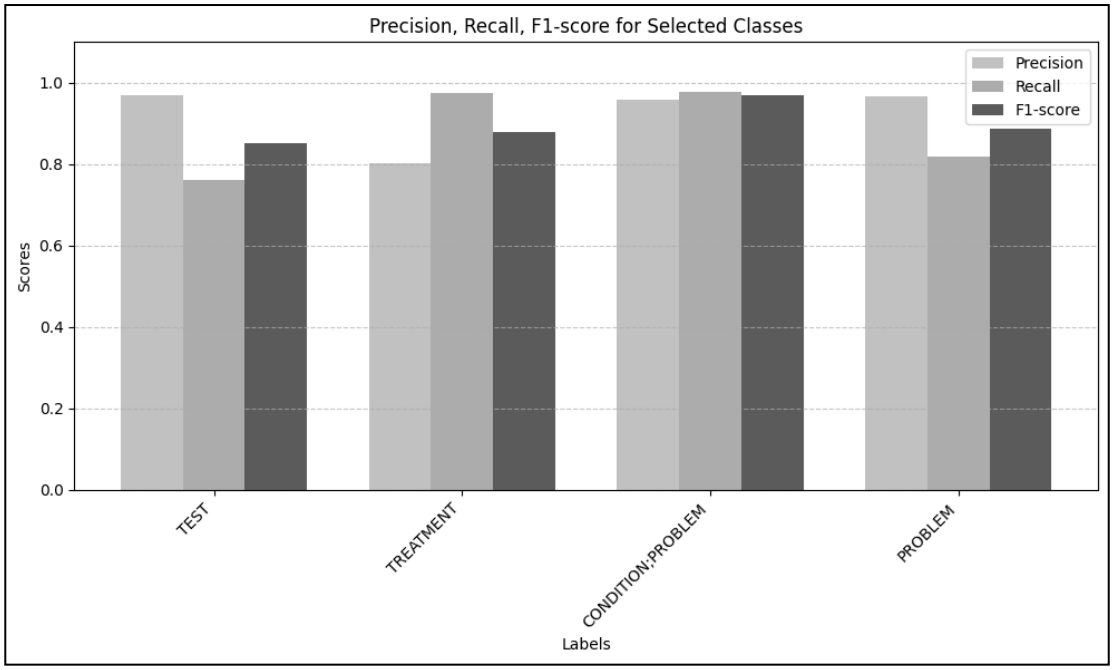

Figure 4.4. Pictorial Representation of Results of Dataset 4

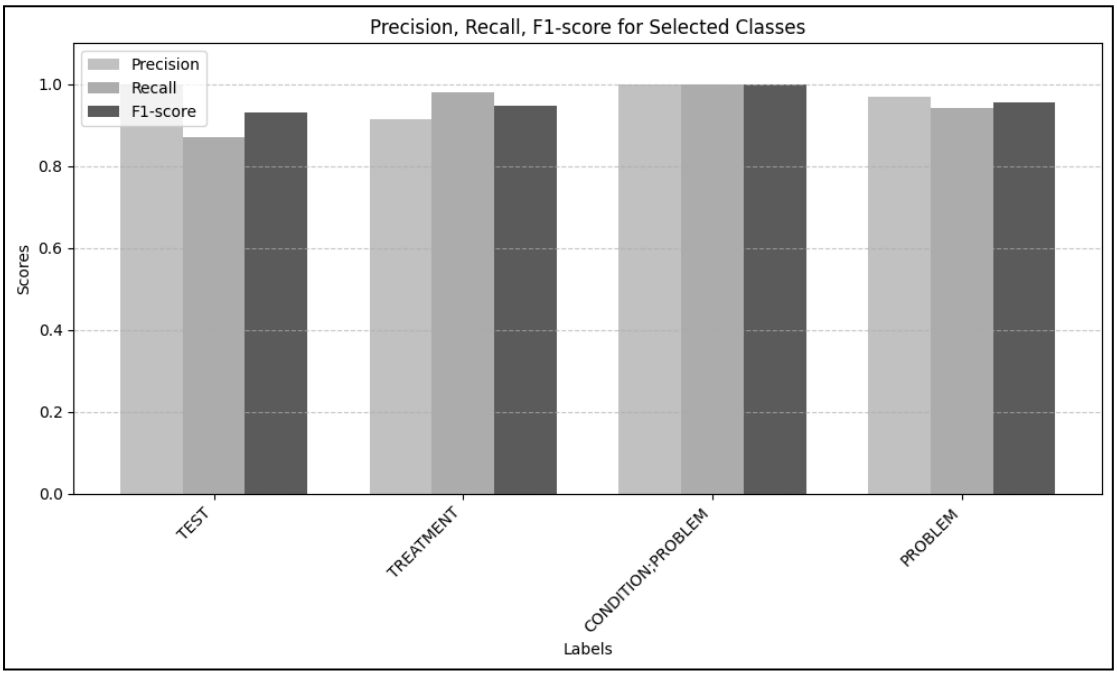

Figure 4.5. Pictorial Representation of Results of Dataset 5

F1-Score represents the proportion of correct predictions made by the model out of the total number of instances. It provides an overall measure of correctness without distinguishing between types of errors. The model achieved F1-Score of 0.91 on Datasets 1 and 2, and 0.94 on Dataset 3, indicating a high level of general reliability across these cases. However, F1-score dropped significantly to 0.88 on Dataset 4, signaling a substantial degradation in performance. This dip suggests that the model

struggled with this particular dataset, possibly due to factors such as noise, class imbalance, or the presence of atypical instances not well-represented in the training data. On Dataset 5, the model reached its highest F1-Score of 0.95, showcasing its potential to perform exceptionally well under favorable conditions. The overall average F1-Score across all five datasets was 0.921, which points to strong, slightly inconsistent, performance.

Precision, defined as the ratio of correctly predicted positive observations to the total predicted positives, assesses the model's ability to avoid false positives. The model performed consistently well in terms of precision, achieving scores of 0.92, 0.91, and 0.91 on Datasets 1, 2, and 3, respectively. Dataset 4 had a notable spike in precision, which was as high as 0.97 in all datasets. Here we can see that the model was just not accurate in this case, but it was also very conservative in its predictions, labeling a given instance as positive only when it was near 100 per cent certain, which reduced the number of false positives. However, this cautious approach could have resulted in the low recall observed for the same dataset. It had excellent high precision (0.95) and high accuracy, again indicating ideal performance in that test case. The model's strength lies in maintaining correctness amongst its positive predictions, with an average across all datasets being 0.932.

Sensitivity or recall is the proportion of actual positives that are correctly predicted as positive. This provides us with an intuitive view of what the model is capable of, i.e., it can find all relevant instances and none that are not. It shows very similar trends in recall values compared to accuracy. In datasets 1, 2, and 3, the recall values were 0.91, 0.91, and 0.90, respectively, which demonstrates that in all cases the model was able to retrieve most of the actual positives. On the other hand, Dataset 4 had a recall of only 0.82 (to confirm that the model missed a lot of real positive cases in this dataset, which we can infer was due to the cautious strategy that the high precision of the model implied). However, Dataset 5, showing us the recall of 0.95, indicates the ideal performance of the model in the good data characteristics case. The average accuracy and average recall over the five datasets was 0.898.

All in all, the results indicate that the model works well across most scenarios except for some tough datasets like Dataset 4. The implication is that this model is especially good at avoiding false alarms, which is critical in high-stakes applications like medical diagnosis and fraud detection. However, the trade-off observed between precision and

recall in Dataset 4 reveals an opportunity for improvement in achieving a better balance between these metrics. Techniques such as data augmentation, hyperparameter tuning, or incorporating more diverse training examples could help address such weaknesses.

Furthermore, the experimental evaluation shows that the model is mostly robust and accurate, with nearly perfect precision and recall in most of the test cases. Dataset 4 suggests that stress testing models on a variety of data is important, and future work to improve the model's adaptability and generalizability to edge cases and true world complexity is needed. Table 4.3 presents the comparison of the performance metrics of a few representative approaches.

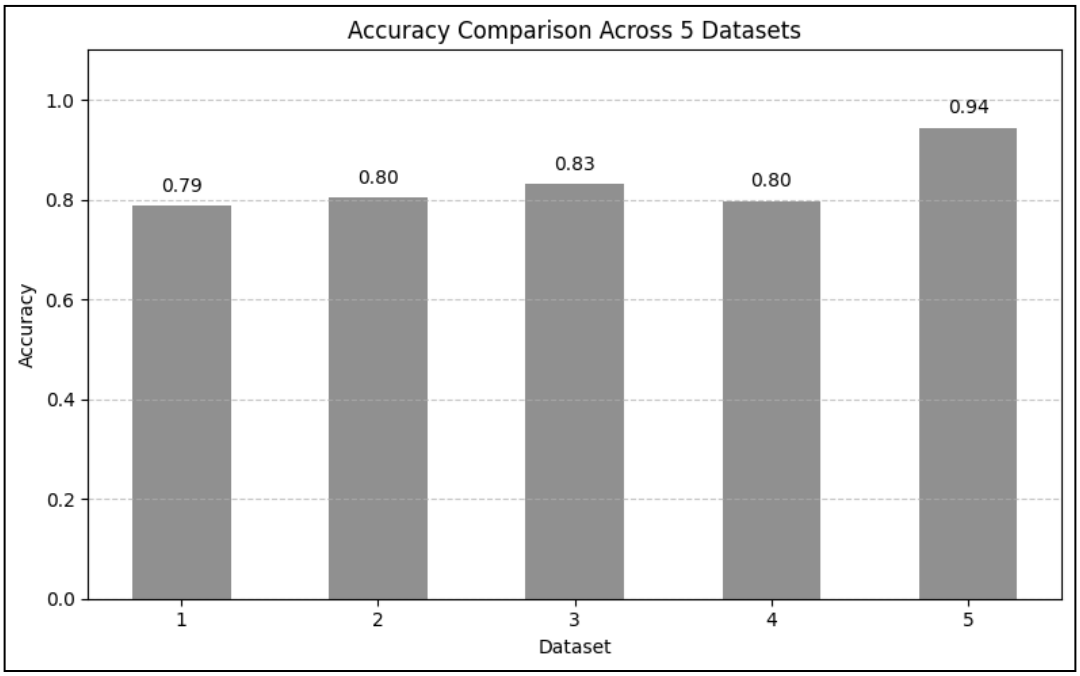

Figure 4.6. Accuracy Comparisons across five Datasets

Table 4.3

Comparison of F1-Score, Precision, and Recall of 5 Datasets of Existing Three Approaches

Paper	Dataset	F1-Score	Precision	Recall
RadGraph: Extracting Clinical Entities and Relations from Radiology Reports [16]	Dataset1	0.701	0.696	0.411
	Dataset2	0.712	0.752	0.454
	Dataset3	0.704	0.723	0.432
	Dataset4	0.711	0.771	0.443
	Dataset5	0.732	0.676	0.410
		0.692		

Classifying social determinants of health from unstructured electronic health records using deep learning-based natural language processing [20]	Dataset1	0.711	0.657	0.662
	Dataset2	0.732	0.725	0.729
	Dataset3	0.752	0.712	0.719
	Dataset4	0.723	0.735	0.742
	Dataset5	0.733	0.713	0.718
		0.745		
Local–Global Memory Neural Network for Medication Prediction [21]	Dataset1	0.686	0.657	0.662
	Dataset2	0.676	0.725	0.729
	Dataset3	0.652	0.712	0.719
	Dataset4	0.677	0.735	0.712
	Dataset5	0.645	0.725	0.729

Chapter 5

Conclusion and Future Work

5. Conclusion and Future Work

The goal of this research was to build a deep learning system that would extract useful information from EHRs and to find accurate and reliable clinical entities and relationships in these. The system was developed using neural network architectures implemented in a Google Colab environment using Python libraries, namely TensorFlow, Keras, and NumPy, which are able to accurately process and handle unstructured medical text.

Experimental evaluation on five data sets showed that the model performs well in terms of accuracy, precision, and recall. Given the overall average accuracy of 0.898, precision of 0.932, and recall of 0.898, the model is seen to be robust and effective in identifying relevant medical entities while producing minimum false positives and false negatives. The model was fruitful, particularly on Datasets 1, 2, 3, and 5, and on Dataset 5, the model scaled to the peak performance of 0.95 accuracy and recall.

We observe their performance degrading on Dataset 4, where the accuracy and recall drop to 0.82, indicating that generalization is a problem with noisier data, class imbalance, and/or counterfactual patterns we haven’t been exposed to before. While it attains a very high precision of 0.97 on the same dataset, the recall trade-off highlights the conservative prediction decisions of the model in such complex scenarios. This indicates an imbalance in the model, which, while great in capturing things the model can say with certainty, can underpredict in other cases where some reasonable amount of uncertainty exists, leading the model to ignore important clinical information.

Results show the comprehensibility and applicability of deep learning for extracting medical information, yet emphasize the need for further augmentation. Future work may entail dealing with these handicaps via practices such as data augmentation, improved preprocessing, hyperparameter optimization, or inclusion of domain-specific knowledge. Furthermore, the same model can also be further developed by expanding the dataset diversity and integrating the real-world clinical data to handle variability and edge cases better.

Finally, the system devised in this work indicates potential for providing automated analysis for EHRs, leading to enhanced clinical decision-making, medical research, and patient monitoring. With further development and validation such models would be essential tools in the modern management of healthcare data.

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